When a drop has to choose: adhesion, capillarity, and the physics of wet contact

A tree frog is hard to peel off a vertical pane of glass. A gecko of about 50 g outperforms it on the same glass, dry: a tokay carries well over forty times its own body weight with nothing between its feet and the pane (Autumn et al., 2000). Both pack pull-off force into a tiny contact patch. On the wet side of that dichotomy, surface tension and short-range molecular adhesion both pull the surfaces back together, while viscosity resists the opening motion and, on fast pull-off, its suction can dominate either attractive term (Maugis, 1992; Stefan, 1874). Which term sets the pull-off force depends on the film thickness, the substrate compliance, and the withdrawal speed. A predictive model covering the crossover between capillary bridge, viscous suction, and molecular adhesion is not known. In this Master's-year project you will derive the limiting scaling laws from first principles, run axisymmetric direct numerical simulations of a pinned liquid bridge with solid–solid adhesion closed by a disjoining pressure, and place a curated set of biological attachment systems on the resulting regime map as quantitative markers from their published rheology and elastic data. Non-Newtonian mucus, compliant substrates, and the full four-group parameter sweep remain as follow-on work beyond this year.

TL;DR

On pull-off, surface tension and molecular adhesion both pull the surfaces back together; viscosity resists the opening motion and, at fast pull-off, its suction can dominate either attractive term. The Master's year takes one tractable slice of the full parameter space: the $(\mathrm{Ca}, \mathrm{Ad})$ plane at fixed gap-to-radius ratio h/R and fixed Tabor–Maugis stiffness μ_T, with $\mathrm{Ca} = \eta U/\gamma$ and $\mathrm{Ad} = W_{\mathrm{adh}}/\gamma$. The JKR limit (crack peel at soft contacts) and the DMT limit (uniform adhesive layer at stiff contacts) enter as asymptotic overlays of the Maugis formulation of sphere adhesion (Maugis, 1992). You will derive the limiting scaling laws, run axisymmetric direct numerical simulations of a liquid bridge with a pinned contact line and a disjoining-pressure closure for solid–solid adhesion, sweep the $(\mathrm{Ca}, \mathrm{Ad})$ plane, and place chameleon tongue, torrent-frog toe pad, clingfish suction, octopus sucker, and insect tarsus on the map as quantitative markers. Non-Newtonian mucus (Weissenberg Wi), compliant pads (elastocapillary compliance $\gamma/(ER)$), and the full $(h/R, \mu_T)$ sweep remain as follow-on axes beyond this year.

Description

Why is a tree frog so hard to peel off a vertical pane of glass? Next door, a gecko outdoes the frog dry: a tokay of about 50 g carries well over forty times its body weight on near-vertical glass, with nothing between its feet and the pane (Autumn et al., 2000). The frog manages a comparable feat by the opposite route — a thin wetting film pinned between toe pad and glass. Two different physics, the same qualitative outcome. This project is about the wet side of that dichotomy: what sets the force to pull two wet surfaces apart when there is no muscle and no active suction pump, only surface tension, molecular adhesion, and viscosity.

As a model experiment, we take two smooth solids held across a thin wetting film, pinned at a prescribed contact radius, and withdraw one at a controlled speed. Three terms set the opposing force on pull-off. The liquid bridge between the surfaces carries a negative Laplace pressure in its neck and contributes a capillary pull-back of order γR, where γ is surface tension and R the contact

radius. Opening the gap at rate $\dot{h}$ generates a Stefan-like viscous suction $F_{\mathrm{visc}} \sim \eta R^4 \dot{h}/h^3$ that resists the motion (Stefan, 1874). If the film is thin enough for molecular forces to bridge the gap, solid–solid attraction contributes $F_{\mathrm{vdW}} \sim AR/D^2$, set by the Hamaker constant A and the residual separation D. However, a predictive model that locates the crossover between these three terms — for a given fluid, geometry, and pull-off speed — is not known.

For this process, the relevant parameter space is at least four-dimensional. A capillary number $\mathrm{Ca} = \eta U/\gamma$ and an adhesion number $\mathrm{Ad} = W_{\mathrm{adh}}/\gamma$ set the two obvious axes. The gap-to-radius ratio h/R enters because the capillary-to-viscous crossover of a pendular bridge has the form $\mathrm{Ca}_\times \sim (h/R)^3 \cos\theta$, so the regime diagram moves materially with geometry: at $h/R = 10^{-2}$ the crossover sits eight orders of magnitude below its value at $h/R = 1$. A stiffness ratio — the Tabor–Maugis number $\mu_T = [RW_{\mathrm{adh}}^2/(E^{*2} z_0^3)]^{1/3}$ (Maugis, 1992) — separates soft JKR contacts, where pull-off is set by crack propagation at the edge of the contact patch, from stiff DMT contacts, where pull-off is a uniform force balance across the whole patch; the same W_{adh} on two substrates of different E^* sits on opposite sides of this transition. For non-Newtonian exemplars a Weissenberg number $\mathrm{Wi} = \dot{\gamma}\tau$ enters, and for compliant substrates an elastocapillary compliance $\gamma/(ER)$ enters.

The Master's year does not map the full four-dimensional space. It takes one tractable slice: the $(\mathrm{Ca}, \mathrm{Ad})$ plane at fixed h/R and μ_T, with the JKR/DMT transition entering as an analytical overlay. Non-Newtonian mucus, compliant pads, and sweeps in $(h/R, \mu_T)$ remain as follow-on work.

Biology occupies distinguishable points in that $(\mathrm{Ca}, \mathrm{Ad})$ slice, and places itself through independently measured rheology and elastic data. The chameleon launches its tongue at prey in ~ 20 ms with a viscoelastic mucus film, in a regime where viscous adhesion dominates over hydrostatic suction; the Brau et al. analysis places the system at $\mathrm{Ca} \approx 6$ with a pronounced Weissenberg contribution (Brau et al., 2016). Frog tongues operate on similar principles but slower (Noel et al., 2017). Torrent frogs of the Rhacophoridae use hexagonal toe-pad microchannels to drain excess water at low pull-off speeds, $\mathrm{Ca} \sim 10^{-5}$–10^{-3}, so that the residual film mediates boundary friction rather than forming a capillary trap (Barnes, 2007; Drotlef et al., 2015). Clingfish and remora attach through sustained hydrostatic underpressures of 20–50 kPa beneath a sealed disc (Wainwright et al., 2013); octopus suckers reach still stronger underpressures, approaching 270 kPa, through active muscle-driven volume change in the acetabulum (Smith, 1991). Insect tarsi combine capillary bridges and viscous pinning with compliant-pad mechanics (Dirks and Federle, 2011; Federle, 2006). The gecko stays off this wet map: its attachment is dry and anisotropic (Autumn et al., 2000), a scalar W_{adh} does not describe it, and it sits at the degenerate $\mathrm{Ca} = 0$ axis.

Placeholder for biology montage: chameleon tongue projection, torrent-frog toe pad (SEM), clingfish ventral disc, octopus sucker acetabulum, insect tarsus. See in-file comments for sources.

Figure 1: Curated biological attachment systems placed on the wet-contact regime map from independently measured rheology and elastic data: chameleon tongue (high Ca, non-Newtonian mucus) (Brau et al., 2016); torrent-frog toe pad (low Ca, small h/R, compliant pad) (Barnes, 2007; Drotlef et al., 2015); clingfish suction (hydrostatic underpressure) (Wainwright et al., 2013); octopus sucker (active-muscle suction) (Smith, 1991); insect tarsus (compliant-pad hairy attachment) (Dirks and Federle, 2011; Federle, 2006).

The idealisation is deliberate: two rigid solids (with a weakly deformable extension for the

compliant-pad cases) bridged by a Newtonian fluid, with solid–solid adhesion closed by a disjoining-pressure potential whose energy integral matches W_{adh}. This closure reproduces the JKR and DMT limits as asymptotes, so the Maugis–Dugdale transition appears as an analytical overlay on the simulation sweep rather than a separately resolved contact-mechanics problem. Non-Newtonian exemplars (tongue mucus) and strongly compliant exemplars (toe pads, tarsal pads) are placed on the resulting map from their published rheology and elastic data; they are not re-simulated at the level of constitutive closure in the baseline DNS campaign. No single published study has swept even this idealised setting systematically with modern two-phase tools.

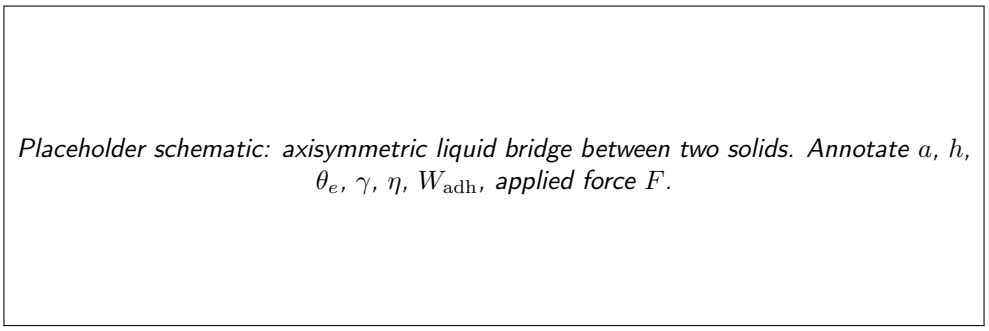


Figure 2: Idealised model problem. Two solids, separated by a wetting liquid bridge, pulled apart at speed U. Surface tension γ, viscosity η, contact angle θ_e (pinned at a prescribed radius a on each solid to avoid moving-contact-line singularities), adhesion energy W_{adh} delivered through a disjoining-pressure potential $\Pi(h)$ with $\int \Pi \, dh = W_{\mathrm{adh}}$. Axisymmetric direct numerical simulations resolve the full flow field, including bridge pinch-off.

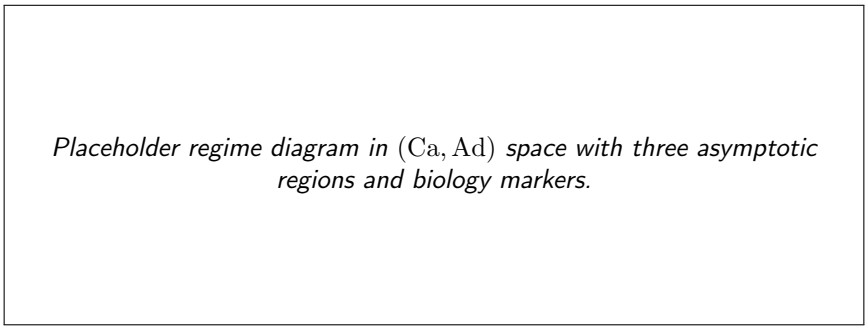


Figure 3: Target of the project: a quantitative regime diagram in $(\mathrm{Ca}, \mathrm{Ad})$ space at fixed h/R and μ_T, separating capillarity-, viscosity-, and adhesion-dominated pull-off. The capillary-to-viscous crossover obeys $\mathrm{Ca}_\times \sim (h/R)^3 \cos\theta$, so the picture moves materially with h/R; the JKR/DMT boundary moves with μ_T. Biological attachment systems from Figure 1 are placed as markers at their respective h/R and μ_T.

What you will do and what you will learn?

The project has four scientific objectives — (i) derive the asymptotic scaling laws and locate the crossover curves; (ii) run a controlled two-phase DNS campaign against those limits; (iii) diagnose where the simple scalings break; (iv) place a curated biological inventory on the resulting map — plus a craft objective (v) of absorbing the CoMPhy-Lab open-source workflow. Items 1–5 below correspond.

1. Derive the limiting scaling laws from first principles: the Laplace capillary bridge, Stefan-type viscous suction on pull-off (Stefan, 1874) with the correct h/R dependence, and the JKR/DMT/Maugis–Dugdale adhesion transition (Maugis, 1992) parametrised by the Tabor–Maugis number μ_T. Locate the crossover curves $\mathrm{Ca}_\times(h/R)$ and the JKR–DMT boundary in μ_T, rather than assuming they collapse onto a single universal two-axis diagram.
2. Set up axisymmetric two-phase direct numerical simulations in Basilisk C (Popinet, 2015) of a liquid bridge between two solids pulled apart at controlled velocity. Pin the contact line at a prescribed radius using Basilisk's contact-angle framework to eliminate the moving-contact-line singularity (no slip model required in the pinned regime); close solid–solid adhesion through a disjoining-pressure potential $\Pi(h)$ with $\int \Pi\, dh = W_{\mathrm{adh}}$; enforce $h_{\min}/\Delta_{\min} \geq 8$ and $h_0/R \geq 0.02$; use $\mathrm{CFL} \leq 0.1$ for the capillary-dominated sweeps. Follow the CoMPhy-Lab two-phase convention (Sanjay, 2022b).
3. Sweep the $(\mathrm{Ca}, \mathrm{Ad})$ plane at fixed h/R and μ_T, measure pull-off force and rupture dynamics, and compare with the analytical limits. Focus on where the simple scalings fail — at the capillary/viscous crossover $\mathrm{Ca}_\times \sim (h/R)^3 \cos\theta$, at the JKR/DMT transition, and where a disjoining closure changes the bridge-rupture geometry relative to a pure-capillary closure. A coarser sweep across h/R and μ_T — together with the non-Newtonian (Wi) and compliant-pad ($\gamma/(ER)$) axes — is left as follow-on work beyond this Master's year.
4. Place a curated set of biological attachment systems on the regime map — chameleon tongue (Brau et al., 2016), frog tongue (Noel et al., 2017), torrent-frog toe pad (Barnes, 2007; Drotlef et al., 2015), clingfish suction (Wainwright et al., 2013), octopus sucker (Smith, 1991), and insect tarsus (Dirks and Federle, 2011; Federle, 2006) — treating the Weissenberg number for the non-Newtonian cases and the elastocapillary compliance $\gamma/(ER)$ for the compliant-pad cases as *placement axes*: systems are positioned on the four-dimensional map using their published rheology and elastic data, with Wi and $\gamma/(ER)$ handled through analytical extensions rather than resimulated at the level of constitutive closure in the baseline DNS campaign. Identify which systems share a regime and which sit at genuinely novel crossovers, and note the gecko as a stated cut: its dry, anisotropic attachment (Autumn et al., 2000) sits at the degenerate $\mathrm{Ca} = 0$ axis and a scalar W_{adh} does not describe it.
5. Learn the full open-source workflow of the CoMPhy Lab: Basilisk setup, adaptive mesh refinement, post-processing in Python, documentation via GitHub, and writing for a physics audience (Sanjay, 2022a,c).

If you have any questions, feel free to contact us vatsal.sanjay@comphy-lab.org/ vatsal.sanjay@durham.ac.uk or drop by Ph255 (Rochester building) at the Department of Physics at Durham University.

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References

Autumn, K., Liang, Y. A., Hsieh, S. T., Zesch, W., Chan, W. P., Kenny, T. W., Fearing, R., and Full, R. J. (2000). "Adhesive Force of a Single Gecko Foot-Hair". *Nature* 405.6787, pp. 681–685. DOI: 10.1038/35015073.

Barnes, W. J. P. (2007). "Functional Morphology and Design Constraints of Smooth Adhesive Pads". *MRS Bull.* 32.6, pp. 479–485. DOI: 10.1557/mrs2007.81.

Brau, F., Lanterbecq, D., Zghikh, L.-N., Bels, V., and Damman, P. (2016). "Dynamics of Prey Prehension by Chameleons through Viscous Adhesion". *Nat. Phys.* 12, pp. 931–935. DOI: 10.1038/nphys3795.

Dirks, J.-H. and Federle, W. (2011). "Fluid-Based Adhesion in Insects – Principles and Challenges". *Soft Matter* 7.23, pp. 11047–11053. DOI: 10.1039/c1sm06269g.

Drotlef, D.-M., Appel, E., Peisker, H., Dening, K., Campo, A. del, Gorb, S. N., and Barnes, W. J. P. (2015). "Morphological Studies of the Toe Pads of the Rock Frog, *Staurois parvus* (family: *Ranidae*) and their Relevance to the Development of New Biomimetically Inspired Reversible Adhesives". *Interface Focus* 5.1, p. 20140036. DOI: 10.1098/rsfs.2014.0036.

Federle, W. (2006). "Why Are So Many Adhesive Pads Hairy?" *J. Exp. Biol.* 209.14, pp. 2611–2621. DOI: 10.1242/jeb.02323.

Maugis, D. (1992). "Adhesion of Spheres: The JKR-DMT Transition Using a Dugdale Model". *J. Colloid Interface Sci.* 150.1, pp. 243–269. DOI: 10.1016/0021-9797(92)90285-T.

Noel, A. C., Guo, H.-Y., Mandica, M., and Hu, D. L. (2017). "Frogs use a viscoelastic tongue and non-Newtonian saliva to catch prey". *J. R. Soc. Interface* 14.127, p. 20160764. DOI: 10.1098/rsif.2016.0764.

Popinet, S. (2015). "A Quadtree-Adaptive Multigrid Solver for the Serre-Green-Naghdi Equations". *J. Comput. Phys.* 302, pp. 336–358. DOI: 10.1016/j.jcp.2015.09.009.

Sanjay, V. (2022a). *Code repository: Drop impact on viscous liquid films*. https://github.com/VatsalSy/Drop-impact-on-viscous-liquid-films (Last accessed: April 1, 2022).

Sanjay, V. (2022b). *Code repository: Impact forces of water drops falling on superhydrophobic surfaces*. https://github.com/VatsalSy/Impact-forces-of-water-drops-falling-on-superhydrophobic-surfaces.git (Last accessed: February 4, 2022).

Sanjay, V. (2022c). *Code repository: When does a drop stop bouncing?* https://github.com/VatsalSy/When-does-a-drop-stop-bouncing (Last accessed: April 20, 2022).

Smith, A. M. (1991). "Negative Pressure Generated by Octopus Suckers: A Study of the Tensile Strength of Water in Nature". *J. Exp. Biol.* 157.1, pp. 257–271. DOI: 10.1242/jeb.157.1.257.

Stefan, J. (1874). "Versuche über die scheinbare Adhäsion". *Sitzungsber. Akad. Wiss. Wien. Math.-Naturwiss. Kl.* 69, pp. 713–735.

Wainwright, D. K., Kleinteich, T., Kleinteich, A., Gorb, S. N., and Summers, A. P. (2013). "Stick Tight: Suction Adhesion on Irregular Surfaces in the Northern Clingfish". *Biol. Lett.* 9.3, p. 20130234. DOI: 10.1098/rsbl.2013.0234.